# Do Now · retrieval

## Lesson 077 · Linear Search: Code and Trace

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Can linear search work with unsorted input?

### Q2

What is not-found sentinel?

### Q3

What does O(n) time mean?

### Q4

Array a=[6,2,9,4]: set a[2]=a[0]+a[1]. Give updated array and total.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Understand — Visit each array value in order; linear search does not require sorted data.

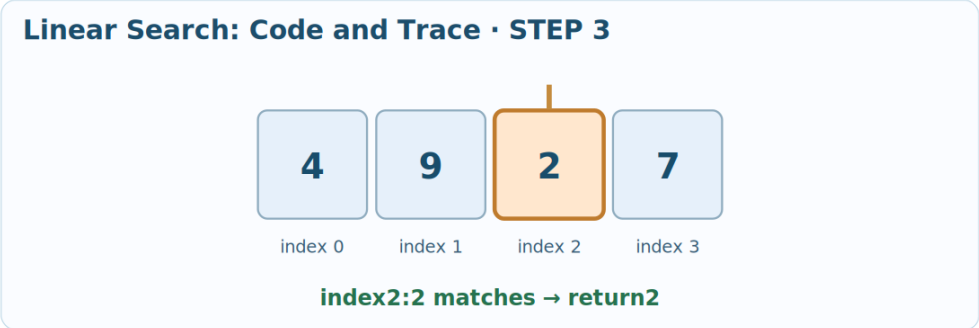
## Core idea 2

Trace — Return the first matching index or a distinct not-found value such as −1.

## Core idea 3

Explain — Worst-case O(n) time, O(1) extra storage; verify first, last, missing and empty inputs.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

linear search — Sequentially compare elements.

sentinel — Special result indicating a condition.

index — Zero-based position.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

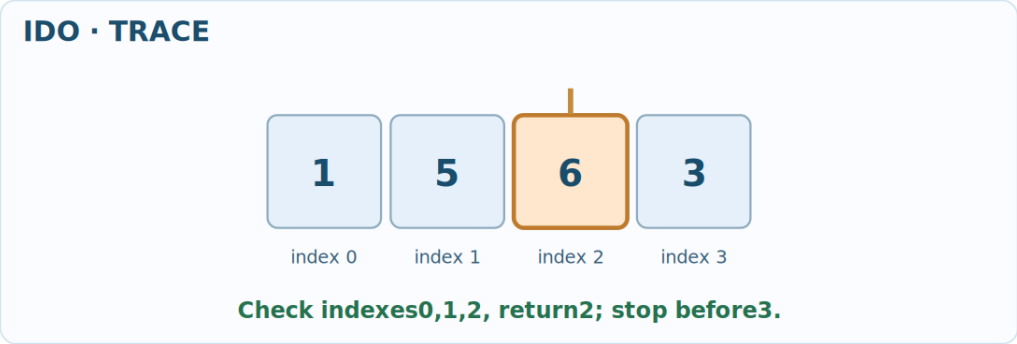
## Problem

Find 6 in unsorted [1,5,6,3].

## Follow the worked method

1. Check the working — Check indexes0,1,2, return2; stop before3.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Check the working
No equal item, so return−1 after four checks.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Trace 2 in [4,9,2,7], give checks and return. [6]

## Question 2

Write C# linear search returning index/−1. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

State linear search best/worst time, space and one edge test. [6]

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

Linear search requires a pre-sorted array.

## Use key vocabulary from memory

1. Explain linear search in your own words.

2. Explain sentinel in your own words.

3. Explain index in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.